TROY K. TSUBOTA

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EDUCATION

Harvard University — **September 2025–present**
Ph.D. Physics — Cambridge, MA

University of California, Berkeley — **August 2021–May 2025**
B.A. Physics (Honors), B.A. Computer Science, B.A. Applied Mathematics (concentration: Probability and Statistics) — Berkeley, CA

ACADEMIC EXPERIENCE

Undergraduate Researcher — **December 2021–August 2025**
UC Berkeley Department of Physics — Berkeley, CA

- Quantum sensing (theoretical) – Prof. Joel Moore
- Pattern formation (theoretical/computational) – Prof. Edgar Knobloch
- Planetary atmospheres (observational) – Dr. Mike Wong

Undergraduate Student Instructor (Teaching Assistant) — **August 2023–December 2024**
UC Berkeley Department of Mathematics — Berkeley, CA

- Fall 2024 – Math 54 – Linear Algebra and Differential Equations – Prof. Zvezdelina Stankova
- Spring 2024 – Math 1B – Calculus II – Prof. Alexander Paulin
- Fall 2023 – Math 1B – Calculus II – Prof. Mark Haiman

Research Intern — **May 2023–August 2023**
Purdue University Department of Physics and Astronomy — West Lafayette, IN

- Quantum sensing (experimental/computational) – Prof. Tongcang Li

PUBLICATIONS

1. **Troy K. Tsubota**, Michael H. Wong, Tom Stallard, Xi Zhang, and Amy A. Simon (2025). *Nature Astronomy* **9** (2), 221–225. [Press release]
2. **Troy Tsubota**, Chang Liu, Benjamin Foster, and Edgar Knobloch (2024). *Physical Review E* **109**, 044210.
3. A. James, P.G.J. Irwin, J. Dobinson, M.H. Wong, **Troy K. Tsubota**, A.A. Simon et al. (2023). *Journal of Geophysical Research: Planets* **128**, e2023JE007904.

PRESENTATIONS

1. Dynamics Days, Davis, CA, January 2024 (poster).
2. AGU Fall Meeting, Magnetospheres in the Outer Solar System Session, San Francisco, CA, December 2023 (talk).
3. Bay Area Planetary Science Conference, Santa Cruz, CA, September 2023 (invited poster).

INDUSTRY EXPERIENCE

AI/ML Engineer — **January 2025–May 2025**
Stealth startup – architecture, engineering, and construction (AEC) — Berkeley, CA

Research Intern – Hardware Technologies — **May 2024–August 2024**
Apple — Cupertino, CA

HONORS AND AWARDS

1. UC Berkeley Physics Departmental Citation [top graduating senior], Spring 2025.
2. Outstanding Graduate Student Instructor (OGSI) Award, Spring 2024.
3. National Science Foundation Research Experience for Undergraduates (REU) at Purdue University, Summer 2023.
4. UC Berkeley Physics Innovators Initiative (Pi^2) Summer Scholar, Summer 2022.
5. Berkeley Physics Undergraduate Research Scholar (BPURS), Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025.